

# Smart India Hackathon 2026



**Team name:** null pointer

**Team Lead:** Dikshit Rishi Jain

**Institute:** National Institute of  
Technology Karnataka

**Problem Statement Title:** AI-Powered  
Learning Gap & Skill Roadmap System

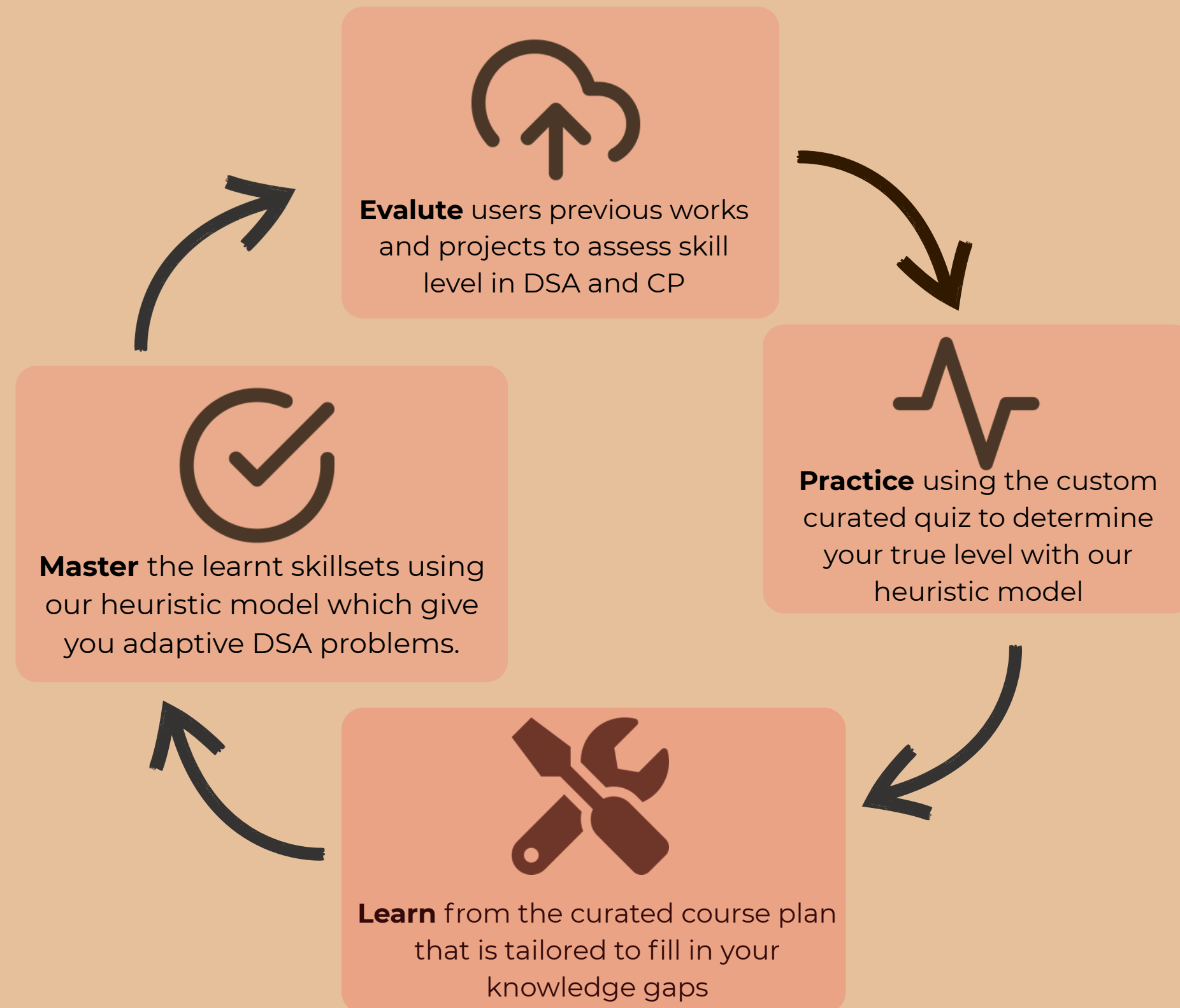
**Theme:** Smart Education

**PS Category:** Software



# TRANSVERSE

DSA and CP at one stop



## WHAT'S Special?



Hybrid dual-engine: LLM curates personalized roadmaps while a deterministic Glicko + vector embedding engine handles exact difficulty, zero hallucination risk.

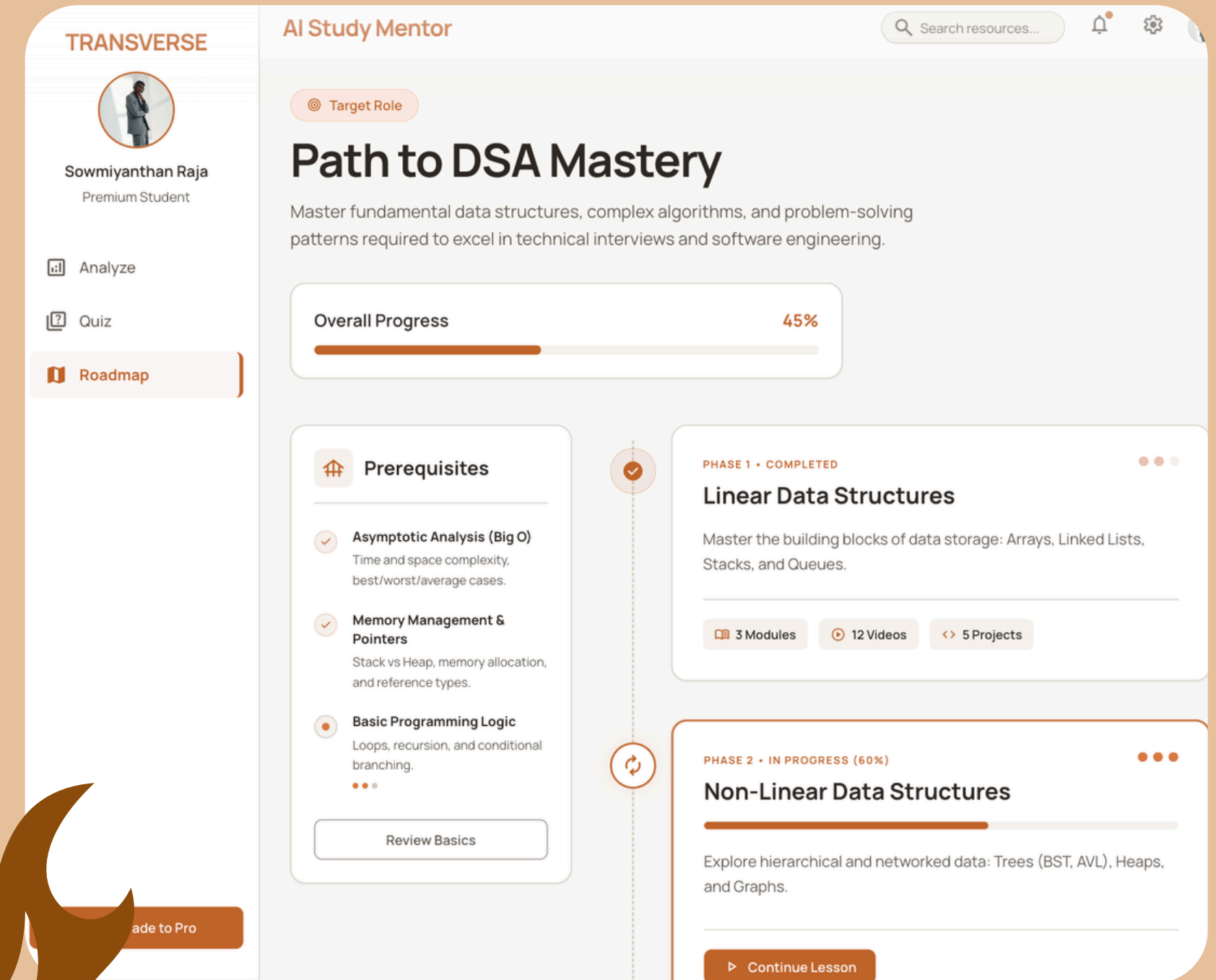
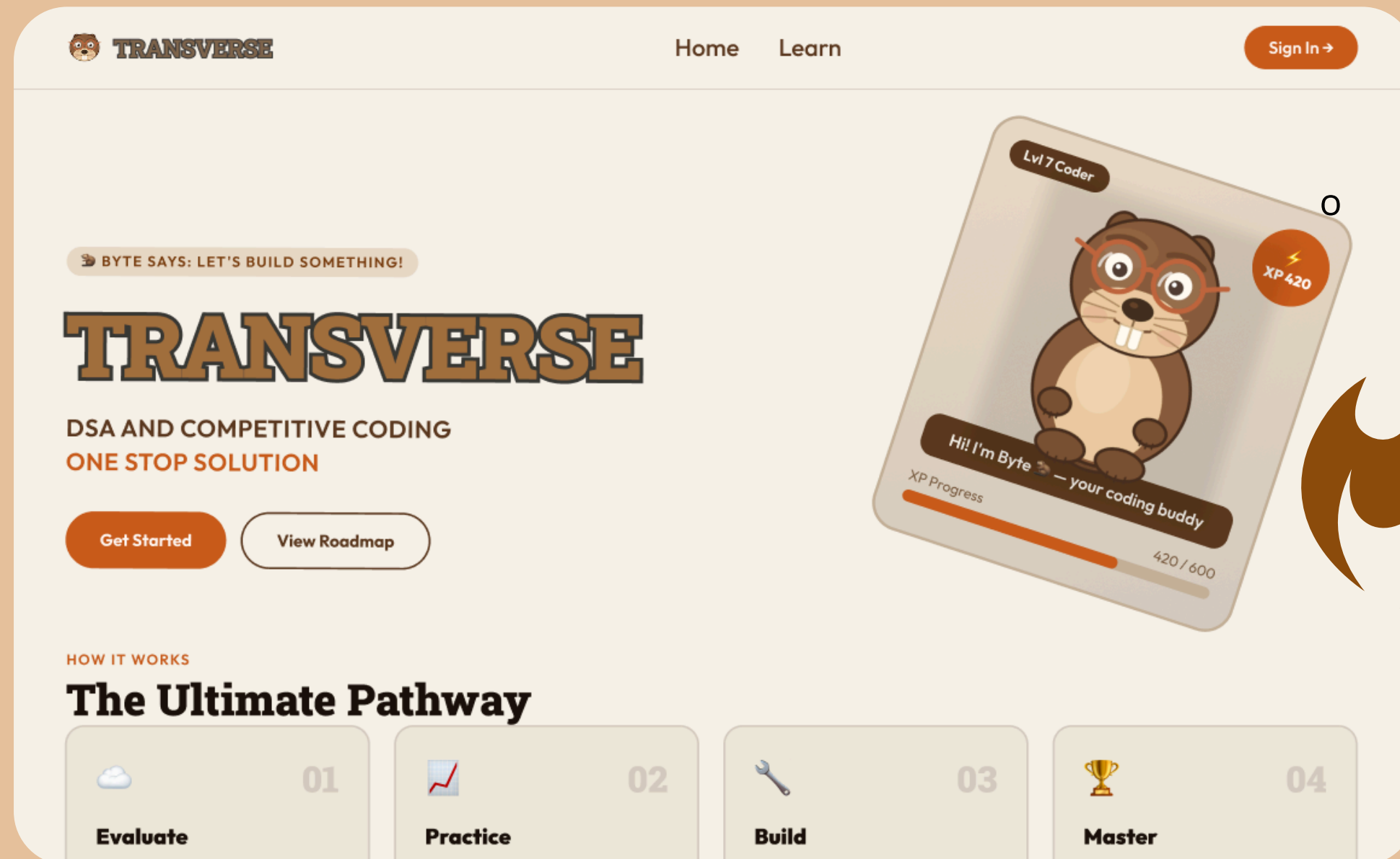
Gamified, Duolingo-style gated progression : phases unlock only once prerequisites are mastered, with streaks and skill badges to drive retention.



Byte, your AI coding mascot, guides you end-to-end; tutorials, compiler, and adaptive quizzes, all on one screen.



# Prototype of the Solution



Sources include Leetcode problems, pre-existing problemsets and tutorials from reputed platforms

# Feasibility

**Judge0 for code execution** — safe, sandboxed code execution runs on an existing open-source engine (self-hosted, resource-limited), not built from scratch.

---

**Grounded, not generative, AI planning.** The study recommender pulls from real already catalogued problem/tutorial databases instead of free generating a roadmap, cutting hallucination risk.

---

**Conversational mascot** as a scoped AI surface — same model, applied to a bounded, low-stakes chat rather than a decision that must be verifiably correct.

# Expansion Opportunities

**Campus-first distribution** - partner with university dev clubs ( IEEE, E-Cells) via free tiers and club leaderboards, turning early adoption into a repeatable channel, not one-off marketing.

---

**Built-in virality** - streaks, shareable challenge results, and skill badges, drive organic growth without needing a paid acquisition budget.

---

**Domain-agnostic engine** - the same roadmap-and-practice architecture can be replayed for other domains: ML, App Dev, Game Dev, Data Science, Web3.

# Tech Stack

## Frontend



## Backend



## DB & Vectors



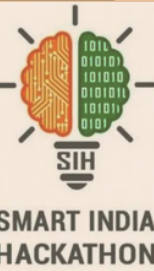
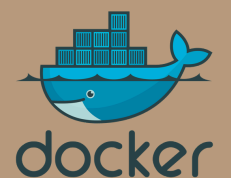
## LLM



## Embedding

BAAI/bge-small-en

## Container





# Major Challenges

## Major challenge 1

Developers are hesitant to pay for a new platform when free YouTube tutorials and LeetCode already exist.

### - SOLUTIONS

- Gamified Duolingo-style mechanics (streaks, gated roadmap, badges) build daily habit that free tools don't offer
- repeated appearance of mascot makes the app standout and **recognizable over time**
- all the same best tools are sourced and used under one site which **makes user learning efficient**



## Benefits to future IT graduates

- Turns prep from guesswork into a measured signal. Students know their actual strengths and gaps from real solving, not a gut feeling about what to study next.
- Shifts what "**progress**" means from a raw problems-solved count to demonstrated understanding.
- Builds **applied engineering intuition**, not just pattern-matching. Mastery requires shipping a small build, not clearing a quiz or a problem count, so the skill transfers to **real interviews and take-home tasks**



## Major challenge 2

A new platform can't compete on library size on day one, and scraping proprietary questions risks legal action.

### - SOLUTIONS

- Launch on top of existing open-access sources (Codeforces API, CSES, NeetCode 150, Striver's A2Z) instead of authoring an original bank from scratch
- The hybrid Glicko + vector-embedding engine is the real differentiator — new questions are seeded via similarity mapping, not raw library size



# The Team



|                           |                            |                                                            |                                                                                                                                 |
|---------------------------|----------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Dikshit Rishi Jain</b> | <b>1<sup>st</sup> year</b> | <b>Btech. in Metallurgical and Materials Engineering</b>   | <b>Dev &amp; Entrepreneur   Co-creator of AlphaJEE (100k+ users)   Co-founder of Meritverse Pvt. Ltd (Software Consultancy)</b> |
| <b>Sowmiyanathan Raja</b> | <b>1<sup>st</sup> year</b> | <b>Btech. in Mining Engineering</b>                        | <b>iOS app developer   Apple Swift Student Challenge winner 2023 and 2024  Author   Brand Owner</b>                             |
| <b>Nilesh Pandey</b>      | <b>1<sup>st</sup> year</b> | <b>Btech. in Metallurgical and Materials Engineering</b>   | <b>Game developer (Unreal Engine, UEFN, Verse)   Unreal Animation Challenge winner 2021</b>                                     |
| <b>Akshay</b>             | <b>1<sup>st</sup> year</b> | <b>Btech. in Mechanical Engineering</b>                    | <b>Web developer   Founder of a studio that builds custom websites for architects.</b>                                          |
| <b>Preet Seth</b>         | <b>1<sup>st</sup> year</b> | <b>Btech. in Mining Engineering</b>                        | <b>Marketing, Strategy &amp; GTM   Chief Marketing for PrepPilot and Crudella (IIT Roorkee startup)</b>                         |
| <b>Bhumi Jain</b>         | <b>1<sup>st</sup> year</b> | <b>Btech. in Electronics and Communication Engineering</b> | <b>Python developer   Smart Brain Contest (IIS Jaipur) winner</b>                                                               |